ISSN 1028-334X, Doklady Earth Sciences, 2022, Vol. 506, Part 2, pp. 710–715. © Pleiades Publishing, Ltd., 2022.
Russian Text © The Author(s), 2022, published in Doklady Rossiiskoi Akademii Nauk. Nauki o Zemle, 2022, Vol. 506, No. 2, pp. 135–141.

GEOLOGY

Concerning Paleogeographic Reconstructions and the State of the Geomagnetic Field at the Precambrian–Paleozoic Boundary: New Paleomagnetic Data for the Olenek Uplift (Siberian Craton)

Corresponding Member of the RAS D. V. Metelkin[a,b,*], E. V. Vinogradov[a,b], V. V. Shcherbakova[c], Academician V. A. Vernikovsky[a,b], S. M. Zakharov[a], and K. E. Nagovitsin[a,b]

Received June 6, 2022; revised June 17, 2022; accepted June 20, 2022

Abstract—We report results of paleomagnetic studies of mafic dikes and sills from the Tas-Yuryakh magmatic complex on the Olenek uplift in the northeast of the Siberian platform. The paleomagnetic record in the rocks corresponds to an episode of anomalous state of the geomagnetic field that persisted from the Ediacaran period (~580 Ma and younger) to the end of the Fortunian age. Paleointensity measurements indicate an extremely low value of the virtual dipole moment during this time. This presumably caused a disruption of the normal Geocentric Axial Dipole model, so much so that the world magnetic anomalies made a substantial contribution. We propose that the Antarctic anomaly influenced the magnetization of the Siberian craton rocks during this period of very low dipole moment. The high latitudes corresponding to the observed paleopole do not correspond to the actual paleogeography of Siberia and can be used for paleoreconstructions only after adjusting for this anomaly. The true position of the Olenek uplift at the Precambrian–Paleozoic boundary was close to 30° S above the southeastern periphery of the African (Tuzo) mantle hot field.

Keywords: paleomagnetism, paleointensity, Thellier method, anomalous geomagnetic field, paleogeographic reconstructions, Ediacaran–early Cambrian, Olenek uplift, Siberian craton

DOI: 10.1134/S1028334X22600414

The issue of the paleogeographic position of the Siberian craton at the Neoproterozoic–Paleozoic boundary is very controversial. The paleomagnetic record stratigraphically corresponding to tillites often yields equatorial latitudes [1]. The presence of glacier deposits in the tropics is confirmed through paleomagnetic data for other cratons as well, which led to the formulation of the "Snowball Earth" hypothesis [2, 3]. However, reliably determined paleomagnetic results also include cases where a high latitude position for Siberia and other cratons is reconstructed. Recent studies of stratigraphically equivalent Ediacaran–early Cambrian rocks of Siberia have systematically shown several coexisting groups of paleomagnetic directions and hyper-frequent reversals in them, as well as other indicators that the geomagnetic field of that time did not correspond to the actualistic Geocentric Axial Dipole model (GAD) [1, 4–6]. The problem of significant differences in coordinates of "coeval" paleomagnetic poles is usually explained by errors in stratigraphy, unreliable age data and/or uncertain magnetization, anomalous plate drift speeds, etc. However, if the configuration of the geomagnetic field was indeed different and if the paleomagnetic data does not correspond to the GAD model, then the tectonic and paleogeographic reconstructions made using this data are incorrect. We assume that one of the causes the field deviated from the normal GAD at the Ediacaran–early Cambrian boundary was an abrupt decrease in the value of the dipole magnetic moment (paleointensity) [7, 8]. This could have led to a drastic increase in the input of anomalous components of the geomagnetic field, which significantly distorted its structure and, respectively, the paleomagnetic record. However, there is currently a shortage evidence for the low intensity of the geomagnetic field and its anomalous configuration. The exact time interval during which this altered state could have existed is also undetermined. The Ediacaran–lower Cambrian sedimentary section of the Olenek uplift has been studied in detail and represents a promising target for confirming these ideas and solving these issues.

[a] *Novosibirsk State University, Novosibirsk, 630090 Russia*
[b] *Trofimuk Institute of Petroleum Geology and Geophysics SB RAS, Novosibirsk, 630090 Russia*
[c] *Borok Geophysical Observatory, department of Schmidt Institute of Physics of the Earth RAS, Borok, Yaroslavl oblast, 152742 Russia*
**e-mail: metelkindv@ipgg.sbras.ru*

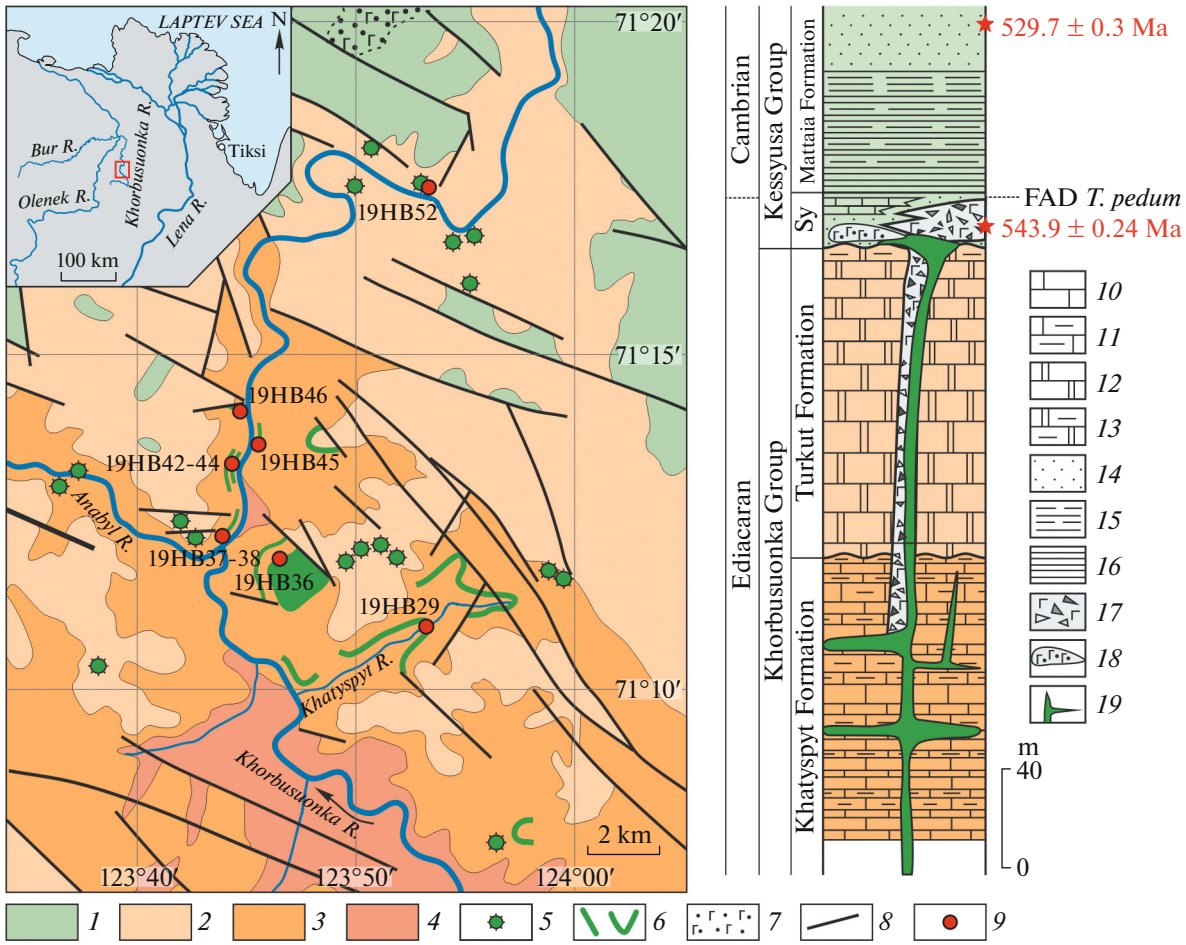


Fig. 1. Geology of the region under study and a section showing the relationships of the Tas-Yukyakh magmatic bodies with sedimentary rocks (modified and supplemented from [14]). (*1*) Ediacaran–early Cambrian deposits of the Kessyusa Gr. and Erkeket Fm.; (*2–4*) Ediacaran Khorbusuonka Gr. deposits: (*2*) Turkut Fm., (*3*) Khatyspyt Fm.; (*4*) Maastakh Fm. and underlying deposits; (*5–7*) early Cambrian Tas-Yuryakh magmatic complex: (*5*) volcanic pipes, (*6*) dikes and sills, (*7*) flows; (*8*) faults; (*9*) paleomagnetic sampling sites with numbers 19HB*; (*10–18*) lithological complexes: (*10*) limestone, (*11*) argillaceous limestone, (*12*) dolomite, (*13*) argillaceous dolomite, (*14*) sandstone, (*15*) intercalating siltstone and sandstone, (*16*) mudstone, (*17*) tuff breccia (in intruding and stratiform bodies), (*18*) flood basalt, (*19*) dolerite (dikes and sills); red stars on the lithological chart show sites of U–Pb age sampling (ref. text); dotted line shows inferred Ediacaran–Cambrian boundary by the first appearance of *Treptichnus pedum* (FAD *T. pedum*) ichnofossils.

In the Khorbusuonka River basin, this section forms a sub-horizontally occurring carbonate-terrigenous sequence consisting of the Khorbusuonka Group (Maastakh, Khatyspyt, and Turkut formations), the Kessyusa Group (Syargalakh, Mattaia, and Chuskun formations), and the Erkeket Formation (Fig. 1) [9]. The age of these rocks has been reliably established by paleontological data and results of chemostratigraphic studies: early Cambrian (not younger than 520 Ma) for the Erkeket Fm., Chuskun Fm., and most of the Mattaia Fm., Ediacaran (not older than 610 Ma) for the underlying part of the Kessyusa Gr. and the entire Khorbusuonka Gr. [9–12]. The Ediacaran rocks host the Tas-Yuryakh magmatic complex that includes multiple dolerite dikes and sills of varying thickness, occasional explosive formations (volcanic pipes), as well as stratiform bodies of breccia and very rare flood basalts [12–14]. The upper stratigraphic occurrence level of the breccias and flood basalts corresponds to the uppermost part of the *Anabarites trisulcatus* zone, which corresponds to the Turkut Fm. and most of the Syargalakh Fm. in the sedimentary section [9, 12]. The U–Pb age of zircon from tuff breccia in one volcanic pipe in the midstream of Khorbusuonka R. is 543.9 ± 0.2 Ma [15]. The age of zircons from thin tuff interbeds in the upper part of the Mattaia Fm., which presumably formed during the final stage of the same magmatic event, is 529.7 ± 0.3 Ma [9, 11]. Therefore, the time range of intrusion of the Khorbusuonka dikes and sills can be narrowed to the Fortunian age, approximately 530 Ma. The geochemical characteristics of the rocks are typical for mantle hot spot magmatism and indicate their connection to intracontinental rifting caused by plume-lithosphere interaction [14].

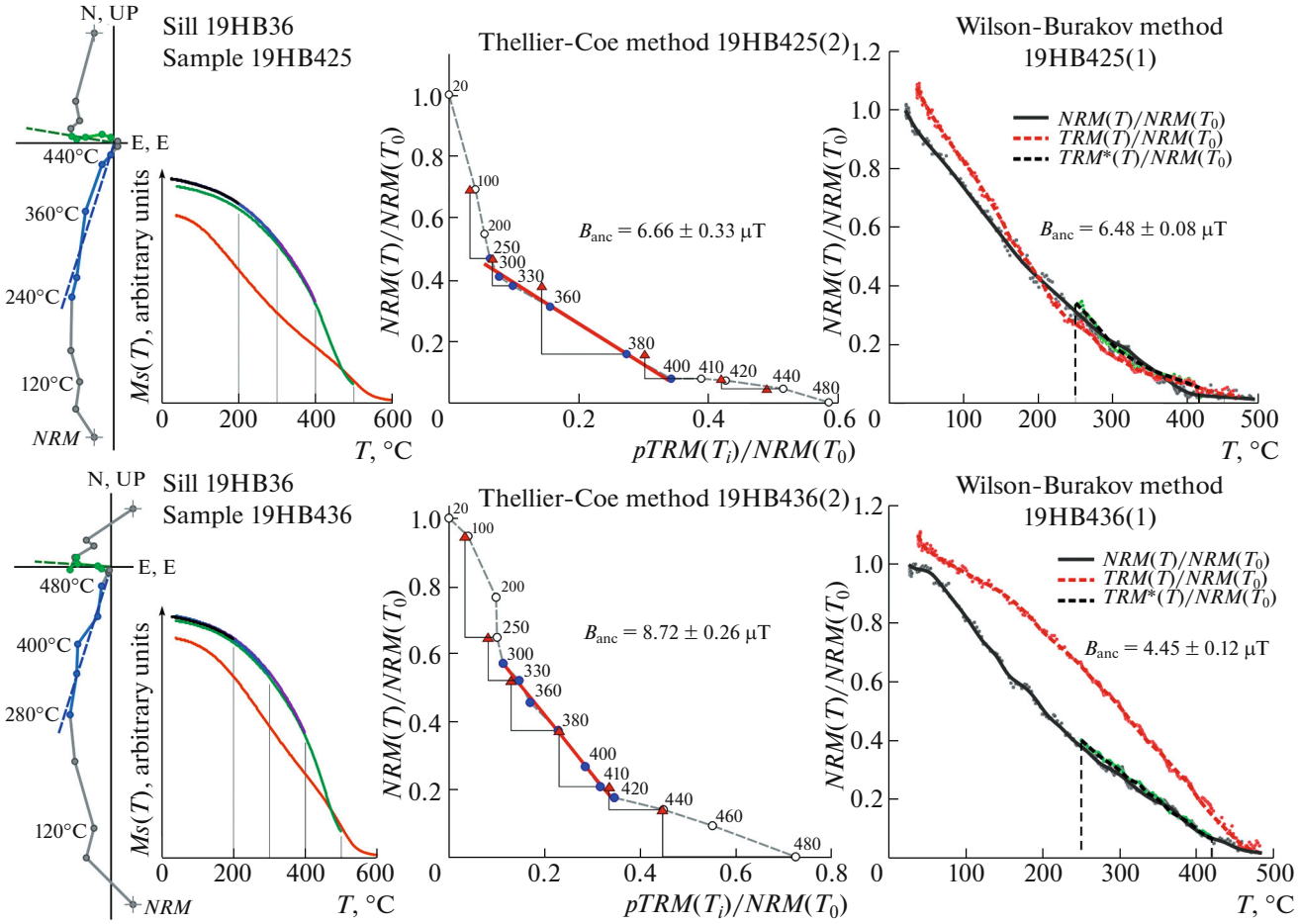


Fig. 2. Typical charts for results of paleomagnetic experiments for two samples and their doubles from sill 19HB36. Left to right—orthogonal Zijderveld plot for incremental thermal demagnetization of natural remanent magnetization (NRM), temperature destruction of saturation magnetization graph ($Ms(T)$) during incremental heating (temperature intervals are shown in different colors); Arai-Nagata diagrams for estimating paleointensity using the Thellier-Coe method; thermomagnetic NRM, TRM, and TRM* curves for estimating paleointensity using the Wilson-Burakov method.

Here we report results of paleomagnetic analysis for nine intrusions (Fig. 1), including measurements of paleointensity. Laboratory work was carried out following standard procedures using the equipment of the Laboratory of Geodynamics and Paleomagnetism of the Central and Eastern Arctic of NSU and the paleomagnetic laboratory of the Borok Geophysical Observatory of IPE RAS.

For almost all samples, thermal stepwise demagnetization in the range 300–450°C yielded a characteristic component of northwestern declination and steep inclination (Fig. 2, Table 1). Analysis of temperature curves of saturation magnetization (Fig. 2) and magnetic susceptibility indicates that the component is stable at temperatures ~420°C. X-ray structure analysis results record a dominant lightly oxidized, high-Fe titanomagnetite in the magnetic fraction. At heating to 500°C for 10 minutes, the *a* parameter of the lattice cell decreases insignificantly, which indicates its single-phase oxidation. Correspondingly, Ms(T) curves show that the value of spontaneous magnetization decreases as the Curie temperature rises (Fig. 2). At the same time, there is an increase of ilmenite content from 7 to 10 wt %, which is probably due to heterophase breakup of the other portion of the primary titanomagnetite.

The position of the figurative dots close to the SD–PSD boundary on the Day-Dunlop plot and the analysis of the domain structure according to the thermomagnetic criterion [16] indicate that the rock is overwhelmingly composed of small pseudo-single domain magnetic grains. In summary, the physical experiments demonstrate that the characteristic magnetization component recorded in titanomagnetite is thermoremanent and primary. This is also confirmed by the results of the baked contact test. The mean magnetization vectors in the dolerite sill (site 19HB42) are not different from those in the Khatyspyt Fm. limestone baked exocontact (site 19HB43). The angular distance between the mean paleomagnetic vectors is

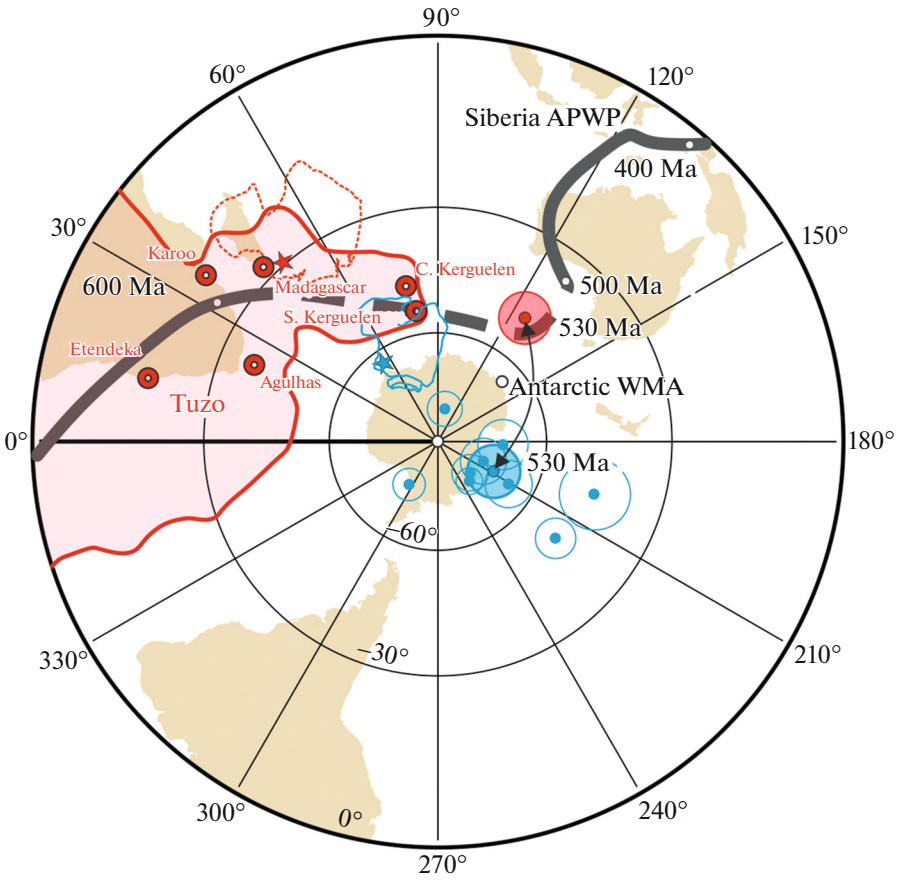


Fig. 3. The Neoproterozoic–early Paleozoic interval of the apparent polar wander path for the Siberian craton from [17] (grey line; dash shows the interpolated Ediacaran–early Cambrian fragment) and the position of the calculated paleopoles. New poles (Table 1) are shown in blue; the corrected mean pole is shown in red. Other key centers and sites discussed in the text are also shown along with the reconstructed paleogeographic position of the Siberian craton (blue—according to the new pole, red dash—after pole correction; stars show the Olenek uplift sampling site) relative to the African mantle hot field (red-white dots in the Tuzo field correspond to positions of main magmatic centers–large igneous provinces). Polar equal angle Wulf net.

$\gamma = 8.2°$ (the critical value being $\gamma c = 15.3°$). At the same time, the angle between the paleomagnetic vector derived from the same dolerite sill site and the one from a nearby Khatyspyt Fm. host limestone outcrop (site 19HB30) is 23.5°, which is markedly higher than $\gamma c = 7.4°$.

The calculated virtual geomagnetic poles are clustered near the eastern coast of Antarctica, far from the apparent polar wander path accepted for paleogeographic reconstructions [17] and imply an unusually high latitude (close to 60° S) for the Siberian craton (Fig. 3). Furthermore, all directions have a positive inclination of the magnetization vector, which is not typical for the mode of frequent geomagnetic field reversals known for the late Precambrian–early Paleozoic period [4, 6]. All of the above leads us to doubt that the obtained paleomagnetic record corresponds to the GAD hypothesis. Therefore, we consider this new data as an addition to the growing system of evidence for a complex configuration of the geomagnetic field at the Ediacaran–Cambrian boundary [5].

We verified the assumption of a low dipole moment by performing experiments to determine the paleointensity (Banc) in 11 dolerite samples from a large sill opposite the mouth of the Anabyl Creek (site 19HB36 on Fig. 1). The 19 determinations by the Thellier-Coe method and 12 determinations by the Wilson-Burakov method satisfy the quality criteria [8]. Most of the Arai-Nagata graphs standard for such a study are non-linear, which is due to secondary low-temperature components (Fig. 2). For Banc estimation, we selected the temperatrure interval corresponding to the deblocking of the characteristic component of magnetization up to the temperature under which the oxidation effect of the primary titanomagnetite was recorded. The average value Banc = 7.58 ± 1.58 μT, corresponds to a virtual dipole moment value of VDM = $1.21 \pm 0.25 \times 10^{22}$ Am^2, which is almost an order of magnitude less than the present-day one. This evidence of ultra-low value of paleointensity during

Table 1. Paleomagnetic directions and coordinates of observed paleopoles

Rock, object, sampling site number	φ	λ	n(S)/N	Paleomagnetic direction				Paleopole		
				D	I	K	α95	Lat	Long	A95
Dolerite, sill, 19HB29	71.18	123.88	13/15	50.5	82.8	85.8	4.5	–75.5	352.7	8.7
Dolerite, sill, 19HB36	71.19	123.77	16/20	285.6	65.2	55.4	5.0	–48.9	219.4	7.4
Dolerite, dike, 19HB37	71.20	123.74	9/10	329.7	72.6	56.6	6.9	–71.8	182.7	11.5
Dolerite, sill, 19HB38	71.20	123.74	10/15	301	57.7	32.5	8.6	–45.8	198.6	10.8
Dolerite, sill, 19HB42	71.21	123.74	9/11	309	73.9	65.7	6.4	–67.1	211.0	11
Hornfels, sill exocontact, 19HB43	71.21	123.74	6/9	306.9	81	13.9	18.6	–73.9	143	35.3
Host limestone for sill 19HB30	71.17	123.82	13/14	327.8	51.4	75.5	4.8	–47.2	165.4	5.4
Dolerite, sill, 19HB44	71.21	123.74	9/11	326.3	76.8	65.7	6.4	–76.1	203.4	11.5
Dolerite, sill, 19HB45	71.23	123.75	8/10	316.5	79.9	225	3.7	–75.8	231.5	6.9
Dolerite, dike, 19HB46	71.24	123.75	16/18	322.1	79.4	53.3	5.1	–77.4	224.1	9.5
Gabbro-dolerite, neck, 19HB52	71.28	123.89	12/12	7.5	75.3	79.4	4.9	–80.6	102.1	8.6
MEAN for intrusions	71.21	123.76	(9)	317.9	75.8	50.1	7.3	–72.3	208.2	12.9

φ, λ–geographic latitude and longitude of sampling site; n(S)/N–number of individual vectors (sampling sites) used in statistics to the total number of studied samples; D–declination; I–inclination; K, α95–Fisher statistic parameters: precision and 95% confidence circle radius; Lat, Long–geographic latitude and longitude, coordinates for the reversed (southern) pole, A95–95% confidence circle.

the Tas-Yuryakh magmatic event is in good agreement with available VDM value estimates for the latest Ediacaran 580–560 Ma [7, 8] and is in fact the first reliable determination for the 530 Ma boundary. Periods of low VDM obviously could have been accompanied by hyper-frequent reversals [4] and alternated with relatively quiet intervals of higher intensity. However, our data indicates that in general such a special field generation mode on the Precambrian–Paleozoic boundary continued for 30–50 Myr, including the entire Fortunian age in addition to a significant part of the Ediacaran period.

The reconstructed drop in VDM could easily have been the cause of disruption of the normal dipole configuration of the geomagnetic field in such a way that the Global Magnetic Anomalies (GMA) and/or large outside sources provided a significant input into its total intensity. Rocks forming close to such sources were magnetized according to the position of the most intense anomalies instead of according to the directions of the dipole magnetic pole (= the geographic pole per the GAD hypothesis). This idea explaining the monopolar character of the determined paleomagnetic record is at the basis of our correction procedure for paleomagnetic directions obtained for the Siberian craton in the identified interval of anomalous state of the geomagnetic field in the Ediacaran–early Cambrian.

According to the available database of paleomagnetic determinations and existing interpolations, during that time the Siberian craton could have been located in the tropical or subtropical zone of the Southern hemisphere [17], at least 20–30 degrees northerly of the latitude obtained for the data on the Tas-Yuryakh complex (Fig. 3). Taking into account the conclusions about the plume nature of the Tas-Yuryakh magmatism [14], as well as the long-term stationarity of the African (Tuzo) hot mantle field [18–20], a probable source of melts could be the plume centers of its southeastern part, which were subsequently responsible for the formation of large igneous provinces of Kerguelen, Madagascar and others (Fig. 3). For such a position of the Siberian craton, the Antarctic WMA was the closest; its center currently being close to the coordinates 65° S, 135° E (ref. Fig. 3). Accepting that this spot was the actual center of the magnetization of the Tas-Yuryakh dolerites, we calculated the Euler rotation (100° clockwise around the pole 74° N, 276° E) needed for the dipole magnetic pole to coincide with it (= the southern geographic pole per the GAD hypothesis). We thus transfer the anomalous coordinates system into the standard one, while the coordinates of the similarly rotated observed pole for the Tas-Yuryakh dolerites (49.1° S, 125.1° E) can be compared with the paleomagnetic poles that correspond to the GAD, i.e. with the standard apparent polar wander path for Siberia. The paleogeographic position of the Olenek uplift corresponding to the adjusted data (Fig. 3) is in full agreement with expected latitudes of about 30° S and does not necessitate abrupt changes in drift kinematics for the Siberian craton during the Ediacaran–early Cambrian.

In conclusion, the analysis performed shows that the anomalous state of the geomagnetic field during the low VDM persisted for at least 30 Myr on the Precambrian–Paleozoic boundary. The high paleolatitudes recorded in the Siberian rocks, despite their formal paleomagnetic reliability, are actually artefacts. They do not reflect the factual paleogeography of the craton and must be corrected to account for the anomalous geomagnetic field configuration. The Antarctic WMA is proposed as the source distorting the paleomagnetic record in the Siberian rocks. This in turn means that this anomaly was stationary for at least the entire Paleozoic era.

FUNDING

This work was financially supported by the Russian Science Foundation (projects 19-17-00091-P, 21-17-00052, and 22-27-00505), and by the Ministry of Science and Higher Education (project no. FSUS-2020-0039).

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

REFERENCES

1. A. V. Shatsillo, S. V. Rud'ko, I. V. Latysheva, et al., Izv., Phys. Solid Earth **56** (6), 833–853 (2020).
2. W. B. Harland, Int. J. Earth Sci. **54**, 45–61 (1964).
3. J. L. Kirschvink, in *The Proterozoic Biosphere: a Multidisciplinary Study* (Cambridge Univ. Press, 1992), pp. 51–52.
4. A. V. Shatsillo, N. B. Kuznetsov, V. E. Pavlov, et al., Dokl. Earth Sci. **465** (2), 1211–1214 (2015).
5. V. E. Pavlov, A. M. Pasenko, A. V. Shatsillo, et al., Izv., Phys. Solid Earth **54** (5), 782–805 (2018).
6. N. L. Dobretsov, D. V. Metelkin, and A. N. Vasilevskiy, Russ. Geol. Geophys. **62** (1), 6–24 (2021).
7. R. K. Bono, J. A. Tarduno, F. Nimmo, and R. D. Cottrell, Nat. Geosci. **12**, 143–147 (2019).
8. V. V. Shcherbakova, V. G. Bakhmutov, D. Thallner, et al., Geophys. J. Int. **220** (3), 1928–1946 (2020).
9. K. Nagovitsin, V. Rogov, V. Marusin, et al., Precambrian Res. **270**, 226–245 (2015).
10. I. A. Vishnevskaya, E. F. Letnikova, N. I. Vetrova, et al., Gondwana Res. **51**, 255–271 (2017).
11. D. V. Grazhdankin, V. V. Marusin, O. P. Izokh, et al., Geol. Mag. **157** (Spec. Iss. 1), 1–13 (2019).
12. V. I. Rogov, G. A. Karlova, V. V. Marusin, B. B. Kochnev, K. E. Nagovitsin, and D. V. Grazhdankin, Russ. Geol. Geophys. **56** (4), 573–583 (2015).
13. B. R. Shpunt and E. A. Shamshina, Dokl. Akad. Nauk SSSR **307** (3), 678–682 (1989).
14. B. B. Kochnev and V. V. Yarmolyuk, et al., Geodyn. Tectonophys. **7**, 233–250 (2016).
15. S. A. Bowring, J. P. Grotzinger, C. E. Isachsen, et al., Science **261**, 1293–1298 (1993).
16. V. V. Shcherbakova, V. P. Shcherbakov, and F. Heider, J. Geophys. Res. **105** (B1) (2000).
17. D. V. Metelkin, V. A. Vernikovsky, and A. Y. Kazansky, Russ. Geol. Geophys. **53** (7), 675–688 (2012).
18. T. H. Torsvik, R. Van der Voo, U. Preeden, et al., Earth-Sci. Rev. **114**, 325–368 (2012).
19. M. I. Kuz'min, V. V. Yarmolyuk, and V. A. Kravchinsky, Geotectonics **45** (6), 415–438 (2011).
20. D. V. Metelkin, V. V. Abashev, V. A. Vernikovsky, and N. E. Mikhaltsov, Dokl. Earth Sci. **501** (2), 1015–1019 (2021).